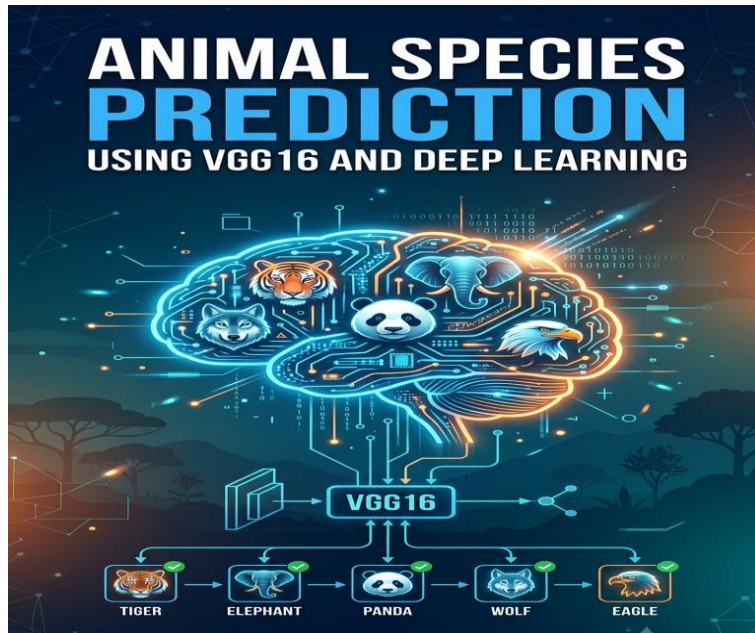


ANIMAL SPECIES PREDICTION USING VGG16 AND DEEP LEARNING



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Source Code Repository: <https://github.com/Prabhatmaurya3239/Animal-Species-Prediction-VGG16>

Submission Date: 01/06/2026

ABSTRACT

Animal species classification is an important task in computer vision and wildlife monitoring systems. Identifying animal species manually from images can be time-consuming and error-prone. This project presents a deep learning-based approach for automatic animal species prediction using the VGG16 Convolutional Neural Network. The model is trained on the Animals-10 dataset, which contains images of ten different animal categories including dog, cat, horse, spider, butterfly, chicken, sheep, cow, squirrel, and elephant. Transfer learning is applied using the pre-trained VGG16 model to improve classification accuracy and reduce training time. Images are preprocessed, resized, and normalized before being fed into the network. The trained model successfully predicts the species of animals from input images with high accuracy. This project demonstrates the effectiveness of deep learning techniques in image classification applications.

1. INTRODUCTION

Computer Vision is a branch of Artificial Intelligence that enables computers to understand and analyze images. Animal species identification plays a significant role in wildlife conservation, biodiversity monitoring, agriculture, and environmental research.

Traditional identification methods require human experts and considerable effort. Deep learning models can automate this process and provide fast and accurate predictions. In this project, a pre-trained VGG16 model is used for classifying animal images into ten categories.

2. PROBLEM STATEMENT

Identifying animal species manually from images is difficult and time-consuming. There is a need for an automated system that can accurately recognize and classify animal species using image data.

3. OBJECTIVE

The main objectives of this project are:

1. To build an image classification model for animal species prediction.
2. To use transfer learning with VGG16.
3. To classify images into ten animal categories.
4. To achieve high prediction accuracy.
5. To demonstrate practical applications of deep learning in computer vision.

4. TECHNOLOGIES USED

- Python
- TensorFlow
- Keras
- NumPy
- Matplotlib
- VGG16
- Kaggle Dataset

5. DATASET DETAILS

- Dataset Name: Animals-10 Dataset
- Source: Kaggle (<https://www.kaggle.com/datasets/alessiocrrado99/animals10/>)
- Total Images: Approximately 26,000+
- Animal Categories:
 1. Dog
 2. Cat
 3. Horse
 4. Spider
 5. Butterfly
 6. Chicken

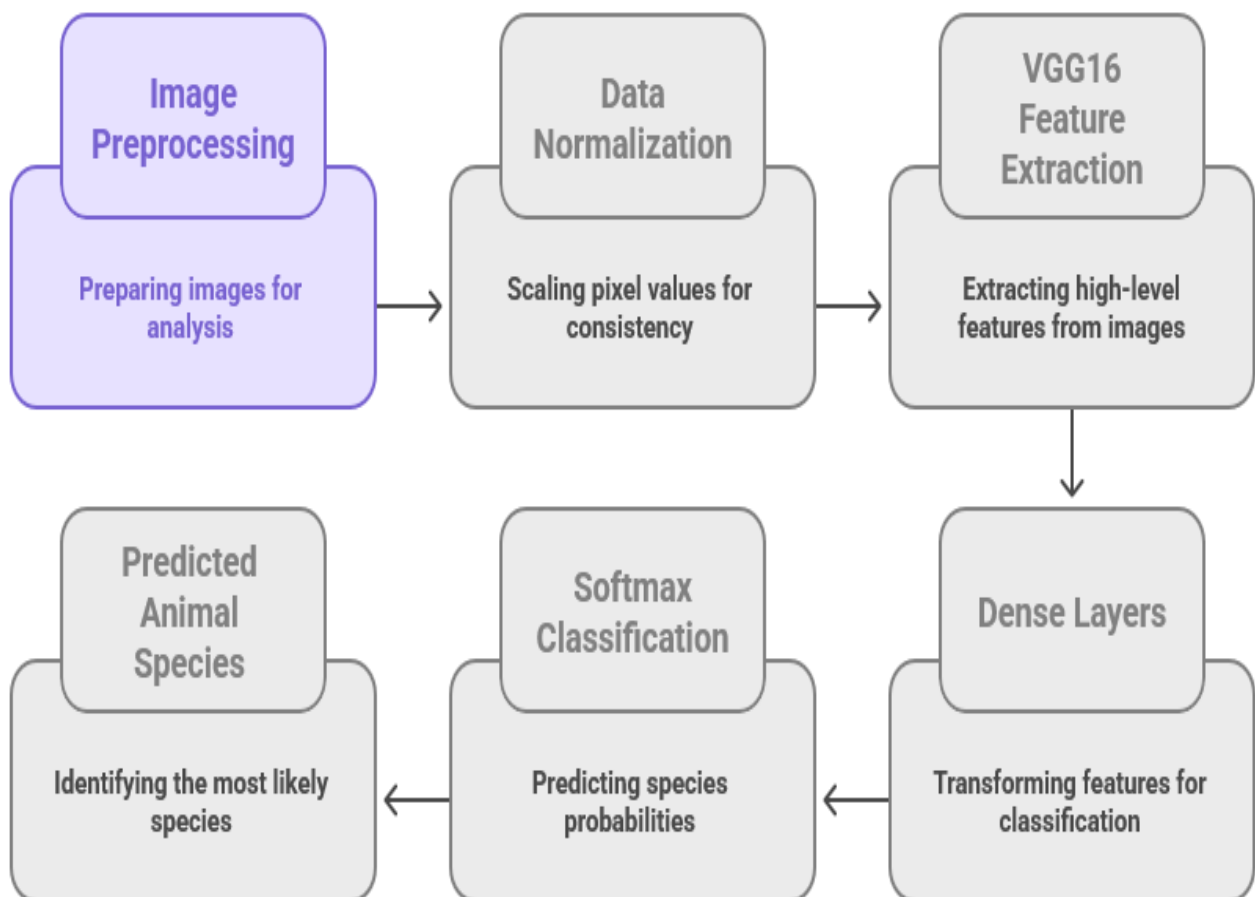
7. Sheep
8. Cow
9. Squirrel
10. Elephant

- Dataset Path:

/kaggle/input/animals10/raw-img

6. SYSTEM ARCHITECTURE

Animal Species Classification Pipeline



7. PROJECT WORKFLOW

Step 1: Dataset Collection

The Animals-10 dataset is downloaded from Kaggle.

Step 2: Data Preprocessing

Images are resized to 224×224 pixels and normalized.

Step 3: Train-Validation Split

80% images are used for training and 20% for validation.

Step 4: Load Pre-trained VGG16

The VGG16 model pre-trained on ImageNet is loaded.

Step 5: Freeze Convolution Layers

Pre-trained layers are frozen to preserve learned features.

Step 6: Add Classification Layers

Dense and Dropout layers are added.

Step 7: Model Compilation

The model is compiled using the Adam optimizer.

Step 8: Model Training

The model is trained on the training dataset.

Step 9: Model Evaluation

Validation accuracy and loss are measured.

Step 10: Prediction

The trained model predicts animal species from new images.

8. MODEL USED

VGG16 is a Convolutional Neural Network developed by the Visual Geometry Group at the University of Oxford.

Reasons for selecting VGG16:

- High classification accuracy
- Pre-trained on ImageNet
- Supports transfer learning
- Easy integration with TensorFlow and Keras
- Reduced training time

9. IMPLEMENTATION

Data Loading:

The dataset is loaded using ImageDataGenerator.

Image Preprocessing:

Images are resized and normalized.

Model Training:

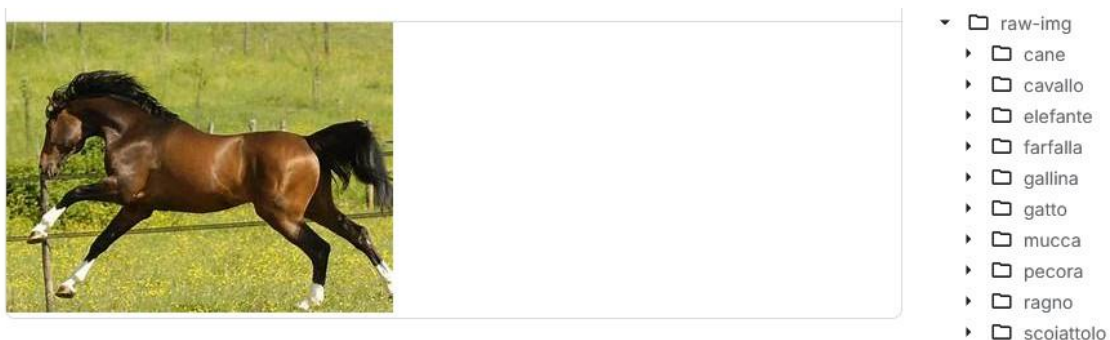
The VGG16 model is trained for multiple epochs.

Prediction:

The model predicts the animal species for unseen images.

Screenshots Included:

1. Dataset Structure Screenshot

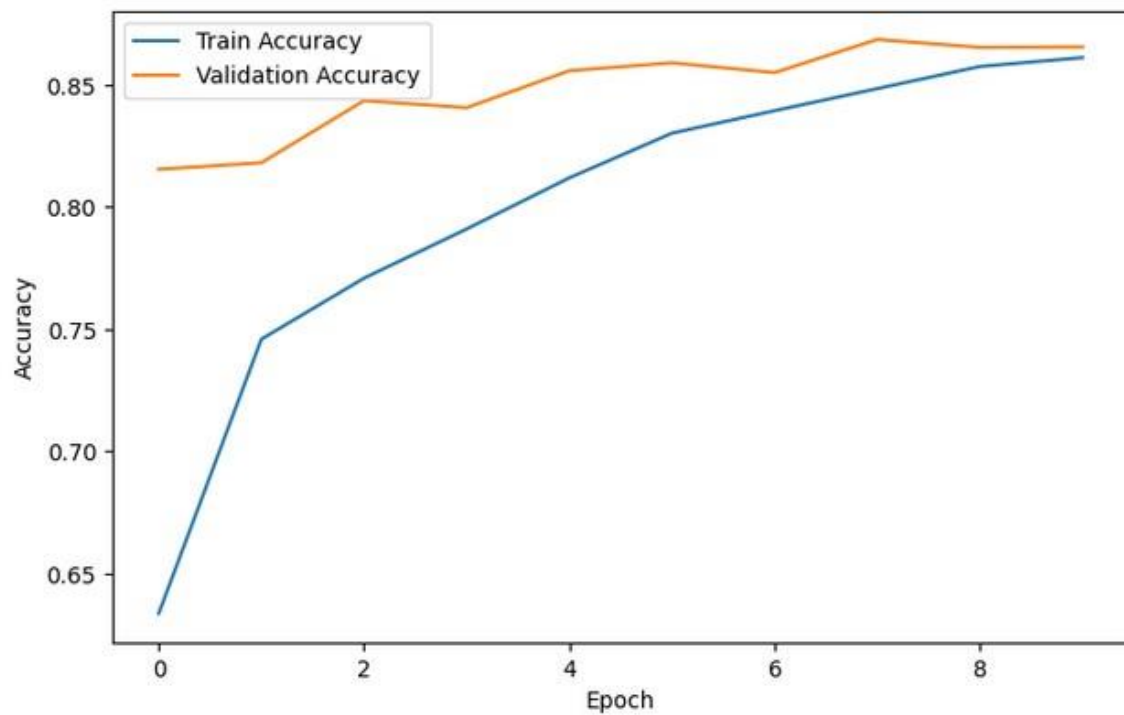


2. Training Process Screenshot

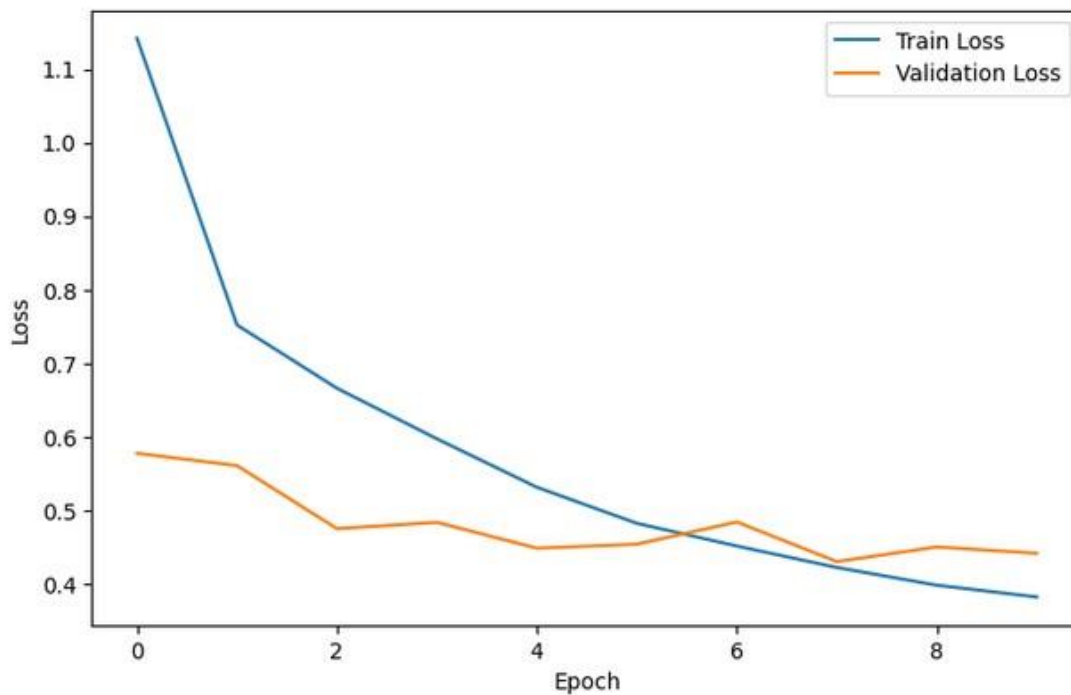
```
history = model.fit(  
    train_data,  
    validation_data=val_data,  
    epochs=10  
)
```

... Epoch 1/10
655/655 — 186s 261ms/step - accuracy: 0.6336 - loss: 1.1422 - val_accuracy: 0.8154 - val_loss: 0.5774
Epoch 2/10
655/655 — 151s 231ms/step - accuracy: 0.7458 - loss: 0.7521 - val_accuracy: 0.8180 - val_loss: 0.5607
Epoch 3/10
655/655 — 152s 232ms/step - accuracy: 0.7708 - loss: 0.6659 - val_accuracy: 0.8435 - val_loss: 0.4748
Epoch 4/10
655/655 — 151s 231ms/step - accuracy: 0.7909 - loss: 0.5971 - val_accuracy: 0.8406 - val_loss: 0.4834
Epoch 5/10
46/655 — 1:54 188ms/step - accuracy: 0.8002 - loss: 0.5140

3. Accuracy Graph Screenshot



4. Loss Graph Screenshot



5. Prediction Output Screenshot

```
from tensorflow.keras.preprocessing import image
import numpy as np

img = image.load_img(
    'e.jpg',
    target_size=(224,224)
)

img_array = image.img_to_array(img)
img_array = np.expand_dims(img_array, axis=0)
img_array = img_array/255.0

prediction = model.predict(img_array)

predicted_class = np.argmax(prediction)

class_names = [
    0: 'Dog',
    1: 'Horse',
    2: 'Elephant', 3: 'Butterfly', 4: 'Chicken', 5: 'Cat', 6: 'Cow', 7: 'Sheep', 8: 'Spider', 9: 'Squirrel'
]

print(class_names[predicted_class])
```

... 1/1 ————— 2s 2s/step
Elephant

10. RESULTS

- ✓ Training Accuracy: 86 %
- ✓ Validation Accuracy: 86 %
- ✓ Sample Prediction:
- ✓ **Input Image:** Elephant
- ✓ **Predicted Output:** Elephant
- ✓ **Performance Analysis:**

The model achieved high classification accuracy and successfully identified animal species from input images.

11. CHALLENGES FACED

1. Large dataset size.
2. High training time.
3. GPU memory limitations.
4. Similar appearance among some animal species.
5. Hyperparameter tuning.
6. Data preprocessing requirements.

12. ADVANTAGES

1. High prediction accuracy.
2. Automated classification.
3. Faster than manual identification.
4. Suitable for real-world applications.
5. Scalable to additional animal species.

13. APPLICATIONS

1. Wildlife Monitoring
2. Biodiversity Research
3. Agriculture
4. Animal Conservation
5. Educational Systems
6. Zoological Studies

14. FUTURE SCOPE

1. Deploy as a web application.
2. Use real-time camera input.
3. Add more animal categories.
4. Improve accuracy using EfficientNet and ResNet.
5. Develop a mobile application.

15. CONCLUSION

This project successfully developed an Animal Species Prediction System using Deep Learning and VGG16. The model was trained on the Animals-10 dataset and achieved high classification accuracy. The system can automatically identify animal species from images and can be extended for various real-world applications such as wildlife conservation, research, and education.

16. REFERENCES

1. Kaggle Animals-10 Dataset <https://www.kaggle.com/datasets/alessiocrrado99/animals10>
2. TensorFlow Documentation <https://www.tensorflow.org/>
3. Keras Documentation <https://keras.io/>
4. VGG16 Research Paper <https://keras.io/api/applications/vgg/>
5. Deep Learning with Python Documentation <https://www.tensorflow.org/tutorials>